

Homework 3 Answers (Questions # and Answers Only)

1. Validity

1(a) Not valid.

1(b) The argument has the invalid form *affirming the consequent*. A counterexample is $p = F$, $q = T$: then $p \rightarrow q$ is true and q is true, but p is false.

1(c) Truth table for $((p \rightarrow q) \wedge q) \rightarrow p$:

p	q	$(p \rightarrow q)$	$(p \rightarrow q) \wedge q$	$((p \rightarrow q) \wedge q) \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

Since the final column is *false* when $p = F$ and $q = T$, the formula is not a tautology, confirming the rule is invalid.

2. Inference

2. Derive $\neg stormy$ from the premises:

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|-----|---------------------------------------|---|
| 1. | $rainy \rightarrow (cold \vee windy)$ | Premise |
| 2. | $\neg cold$ | Premise |
| 3. | $windy \rightarrow uncomfortable$ | Premise |
| 4. | $\neg uncomfortable$ | Premise |
| 5. | $stormy \rightarrow rainy$ | Premise |
| 6. | Assume $stormy$ | Assumption (for contradiction) |
| 7. | $rainy$ | 5, 6 Modus Ponens |
| 8. | $cold \vee windy$ | 1, 7 Modus Ponens |
| 9. | $windy$ | 8, 2 Disjunctive Syllogism |
| 10. | $uncomfortable$ | 3, 9 Modus Ponens |
| 11. | $\perp$ | 10, 4 Contradiction |
| 12. | $\neg stormy$ | 6–11 Negation Introduction (Indirect Proof) |

3. FOL Propositions (Domain: Books)

3(1) $\forall x ((novel(x) \wedge bestselling(x)) \rightarrow (engaging(x) \wedge wellwritten(x)))$.

3(2) $\exists x (textbook(x) \wedge \neg difficult(x))$.

3(3) $(\forall x (biography(x) \rightarrow informative(x))) \wedge (\exists x (novel(x) \wedge bestselling(x)))$.

4. FOL Propositions (Domains: People and Movies)

4(a) $\forall x \exists y (comedy(y) \wedge enjoys(x, y))$.

4(b) $\exists y (drama(y) \wedge \forall x enjoys(x, y))$.

4(c) $\forall x \exists y (comedy(y) \wedge \neg enjoys(x, y))$.

4(d) $\forall x \forall y (directed(x, y) \rightarrow enjoys(x, y))$.

4(e) $\forall y (acclaimed(y) \rightarrow \exists x directed(x, y))$.

5. Negation of FOL Propositions

5(a) $\neg(\forall x (P(x) \rightarrow Q(x))) \equiv \exists x (P(x) \wedge \neg Q(x))$.

5(b) $\neg(\exists x (R(x) \wedge \forall y S(x, y))) \equiv \forall x (\neg R(x) \vee \exists y \neg S(x, y))$.

5(c) $\neg(\forall x (\exists y P(x, y) \rightarrow \forall z Q(z))) \equiv \exists x (\exists y P(x, y) \wedge \exists z \neg Q(z))$.